

Medical Robotics CS4270 - Assignment Submission

Summer 2026

Exercise Sheet 8

Group 8

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Please upload your solutions via the submission activity in the Medical Robotics Moodle course. Submissions must include the names of all group members and be uploaded as a .pdf file.

1. Motion Correlation I

1.1. 1.A

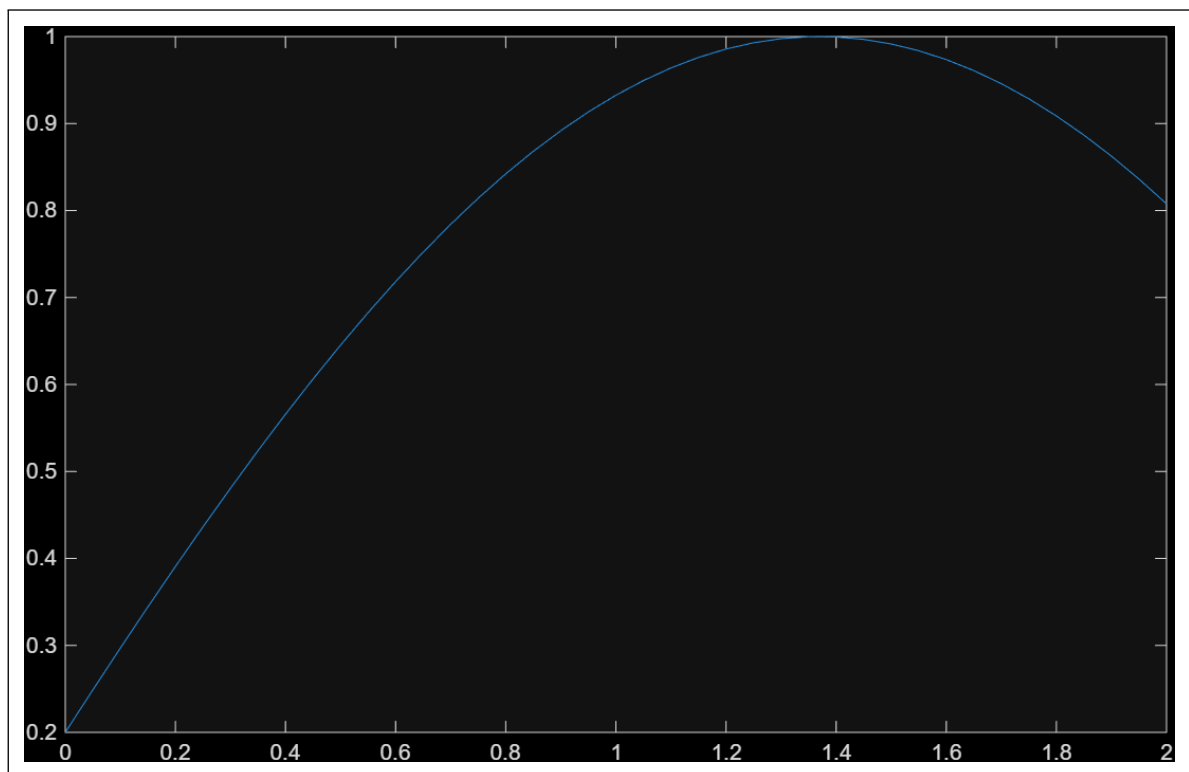


Figure 1: Patient positions plotted over time.

1.2. 1.B - 1.D

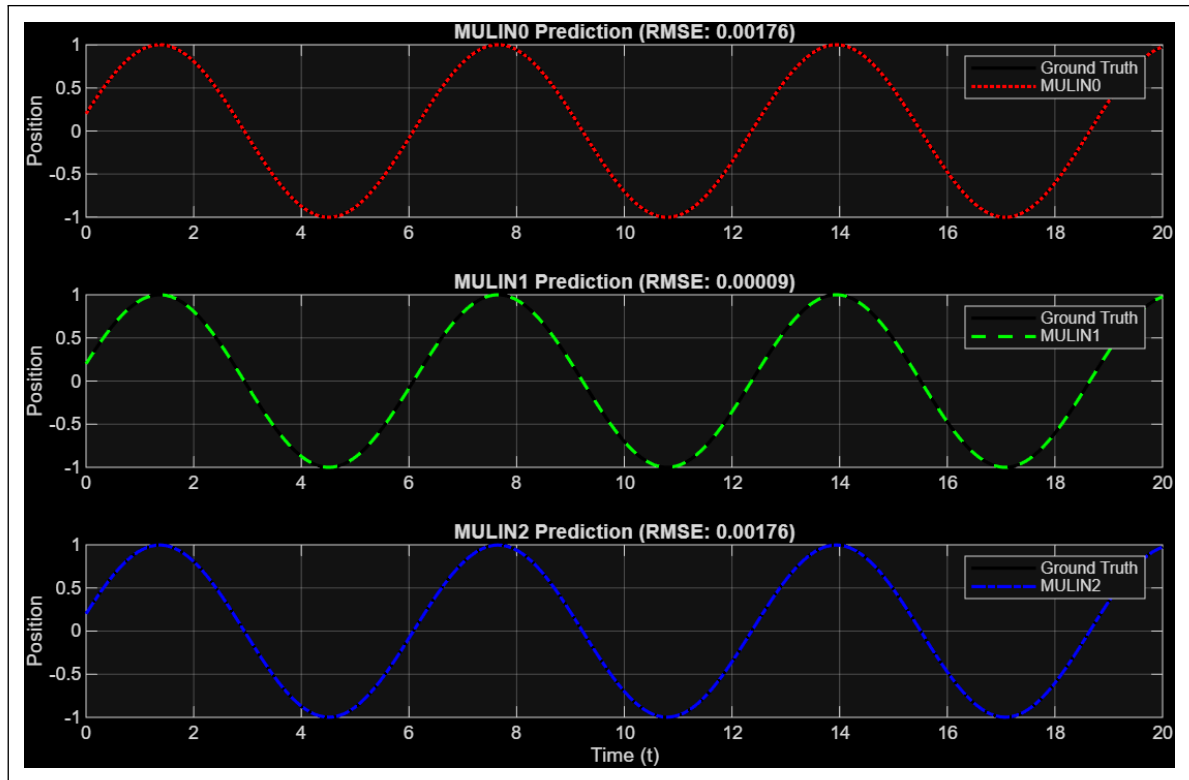


Figure 2: Comparison of MULIN prediction models against ground truth.

1.3. 1.E

Console Output:

RMSE for MULIN0: 0.00176229

RMSE for MULIN1: 0.00008806

RMSE for MULIN2: 0.00176230

Observing the RMSE values, it is clear that the MULIN order 1 is the best for the following case.

2. Least Mean Squares

2.1. 2.A

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Derivations

Inclass sheet

we know that

$$e_n = y_n - \hat{y}_n$$

$$\hat{y}_{n+1} = w_n y_n$$

$$w_{n+1} = w_n + e_n y_n$$

given

$$K=1 \quad y_0 = \frac{\sqrt{2}}{2} \quad y_1 = 1 \quad y_2 = \frac{\sqrt{2}}{2} \quad y_3 = 0$$

let's assume $w_0 = 0$

$\therefore$ At time step $n=0$

$$\hat{y}_1 = w_0 y_0 = 0 \times \frac{\sqrt{2}}{2} = 0$$

$$\therefore e_0 = y_0 - \hat{y}_1$$

$$\therefore e = \frac{\sqrt{2}}{2}$$

$$w_1 = w_0 + e_0 y_0$$

$$\therefore w = 0.5$$

For $n=1$

$$\hat{y}_2 = w_1 y_1 = 0.5 \times 1 = 0.5$$

$$e_1 = y_1 - \hat{y}_1 = 1 - 0.5 = 0.5$$

$$\therefore w_2 = w_1 + e_1 y_1 = 0.5 + (1 \times 1) = 1.5$$

for $n=2$

$$\hat{y}_3 = w_2 y_2 = 1.5 \times \frac{\sqrt{2}}{2} = \frac{3\sqrt{2}}{4}$$

$$e_2 = y_2 - \hat{y}_1 = \frac{\sqrt{2}}{2} - 0.5 = \frac{\sqrt{2}-1}{2}$$

$$w_3 = w_2 + e_2 y_2 = 1.5 + \left(\frac{\sqrt{2}-1}{2} \right) \left(\frac{\sqrt{2}}{2} \right)$$

$$= 2 - \frac{\sqrt{2}}{4}$$

For $n=3$

$$\hat{y}_4 = w_3 y_3$$

$$\therefore y_3 = 0$$

$$\therefore \hat{y}_4 = w_3 \times 0 = 0$$

$$\therefore \underline{\underline{\hat{y}_4 = 0}}$$

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Figure 3: Calculation of $\hat{y}_4$ using the LMS algorithm.

2.2. 2.B

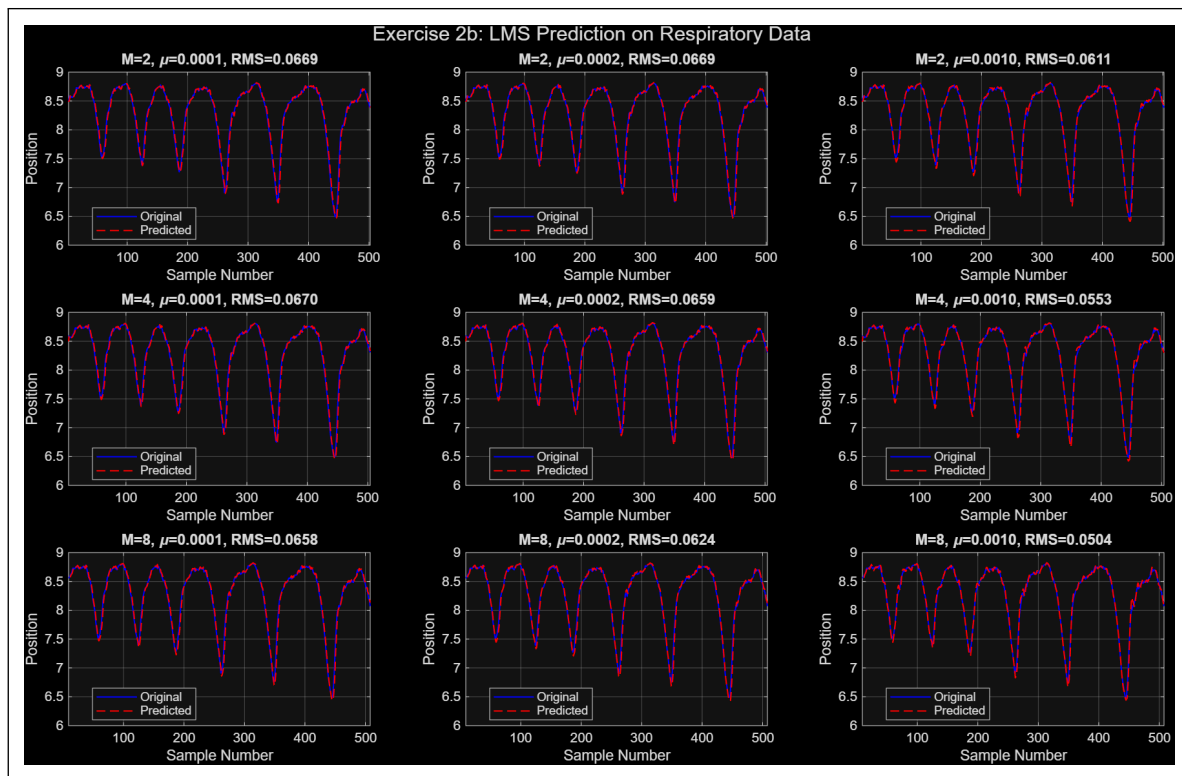


Figure 4: All the Plots wrt y , M and μ .

2.3. 2.C

===== LMS RESULTS =====		
M	mu	RMS
2	0.0001	0.066892
2	0.0002	0.066931
2	0.0010	0.061117
4	0.0001	0.066981
4	0.0002	0.065891
4	0.0010	0.055339
8	0.0001	0.065843
8	0.0002	0.062422
8	0.0010	0.050442

BEST: M=8, mu=0.0010, RMS=0.050442		
=====		

As the console output shows the value of RMSE decreases as the value of M and μ increases, This is due to the algorithm having access to more data points.